

MANHATTAN 6 SSW, KS, USA

WMO: 724540

Lat: 39.1027N

Lon: 96.6098W

Elev: 1137

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	7.8	-5.7	4.2	5.8	-1.8	5.2	8.9	17.6	42.9	16.2	38.6	4.3	N/A	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	98.3	73.8	94.7	74.1	91.5	73.6	78.0	90.4	76.7	89.5	75.3	88.0	8.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	135.0	85.0	73.0	127.6	83.8	71.4	121.0	82.5	42.2	90.9	40.9	89.3	39.6	87.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	14.8	13.1	-5.9	102.3	5.1	3.4	-9.6	104.8	-12.6	106.8	-15.5	108.8	-19.2	111.3		
			-7.0	80.4	5.7	1.7	-11.2	81.6	-14.5	82.6	-17.7	83.5	-21.9	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	30.2	32.9	46.2	55.6	65.1	75.0	78.9	76.7	69.7	56.7	44.5	33.1
	DBStd	19.72	11.90	12.37	11.67	10.11	8.79	6.47	6.59	6.51	8.42	9.90	10.74	10.82
	HDD50	2184	616	492	212	57	5	0	0	0	0	46	228	528
	HDD65	4946	1079	898	591	311	113	5	1	2	47	294	617	988
	CDD50	4192	2	13	93	225	474	749	896	829	592	252	62	5
	CDD65	1482	0	0	8	28	117	304	433	366	189	35	2	0
	CDH74	15871	0	3	79	310	1078	3261	5005	3904	1854	356	21	0
	CDH80	6964	0	0	12	76	335	1375	2521	1831	719	94	1	0
Wind	WSAvg	6.0	5.9	6.2	7.5	7.9	6.3	6.0	4.9	4.6	5.1	5.6	6.0	5.9
Precipitation	PrecAvg	33.4	0.7	1.1	2.0	3.1	4.7	5.0	4.6	4.0	3.0	2.4	1.6	1.2
	PrecMax	51.9	1.6	3.3	4.7	9.3	13.1	10.2	15.5	8.5	5.3	5.6	5.0	3.7
	PrecMin	24.4	0.1	0.1	0.0	0.9	1.1	0.4	1.9	0.7	0.4	0.2	0.0	0.0
	PrecStd	6.3	0.4	0.8	1.2	1.7	3.1	2.3	3.0	1.7	1.3	1.3	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.2	71.3	81.1	87.3	91.1	98.4	103.3	102.0	95.3	88.1	77.4	64.6
		MCWB	51.8	52.1	60.9	65.1	69.3	73.9	73.2	74.6	72.0	68.1	59.3	52.8
	2%	DB	58.2	64.1	75.4	81.7	87.0	94.2	99.4	96.8	91.0	82.3	71.6	58.5
		MCWB	46.3	50.4	57.5	61.8	68.4	74.8	74.0	73.2	72.2	64.5	57.7	48.1
	5%	DB	52.8	58.1	70.5	77.4	83.9	91.1	95.4	93.2	87.7	77.7	66.6	53.9
		MCWB	42.9	48.2	54.9	60.0	67.3	74.3	74.7	73.2	71.2	61.9	53.3	44.9
	10%	DB	47.8	52.4	65.5	73.0	80.4	87.7	91.8	89.5	84.2	73.5	61.6	49.6
		MCWB	40.1	43.3	52.7	58.3	66.3	73.2	74.6	72.3	69.3	59.8	50.6	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	56.7	64.8	66.6	73.5	79.2	80.5	78.5	75.8	71.0	62.5	56.7
		MCDB	58.2	64.0	78.3	80.7	84.5	90.2	92.4	90.7	89.7	80.9	72.1	62.1
	2%	WB	46.6	53.4	61.8	64.5	71.2	77.5	78.3	77.1	74.3	68.0	58.6	50.1
		MCDB	55.7	61.6	70.9	77.2	81.9	90.1	90.7	89.7	87.8	77.0	66.6	56.3
	5%	WB	43.4	48.5	58.1	62.3	69.6	75.8	77.0	75.6	72.9	64.7	55.2	45.4
		MCDB	51.4	60.0	69.1	74.5	81.0	88.0	89.9	88.4	85.6	73.8	65.4	52.9
	10%	WB	39.6	43.3	53.8	59.9	68.2	74.4	75.5	74.3	71.3	61.5	51.8	42.2
		MCDB	47.6	52.5	64.3	70.3	78.8	86.5	88.4	85.8	82.6	70.5	60.1	49.1
Mean Daily Temperature Range	5% DB	MDBR	20.7	21.3	22.6	22.7	21.0	20.6	20.9	21.2	22.5	23.9	22.3	19.4
		MCDBR	28.4	30.2	29.6	27.3	24.3	22.9	24.1	24.6	22.7	27.5	28.3	25.6
		MCWBR	18.0	19.4	16.1	14.0	10.7	9.5	7.7	8.8	9.2	13.9	15.5	16.9
	5% WB	MCDBR	27.3	28.6	25.5	24.3	21.1	21.4	20.3	21.0	20.8	23.4	25.2	23.2
		MCWBR	18.0	19.5	13.8	13.8	10.7	10.6	8.7	9.1	9.2	14.5	16.3	16.7
Clear-Sky Solar Irradiance	taub	0.286	0.301	0.332	0.378	0.408	0.420	0.426	0.425	0.381	0.341	0.300	0.281	
	taud	2.486	2.453	2.413	2.309	2.291	2.292	2.299	2.301	2.376	2.468	2.514	2.503	
	Ebn at Noon	285	294	294	285	277	273	270	267	273	275	277	278	
	Edh at Noon	24	29	34	40	42	42	41	40	35	29	24	23	
All-Sky Solar Radiation	RadAvg	736	981	1284	1629	1860	2109	2078	1856	1534	1121	807	632	
	RadStd	63	100	96	102	149	127	143	93	99	117	67	60	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page